PRODUCT REQUIREMENT DOCUMENT (PRD)

Project Name: ORION - Orchestrated Reimbursement Intelligence Operations Network

Version: 1.0

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Project Overview

- **Problem Statement:** Enterprise expense reimbursement is a process burdened by manual bottlenecks, inconsistent policy enforcement, and fraud vulnerability. Finance teams spend significant hours reviewing each claim, cross-referencing receipts, and enforcing reimbursement policies by hand. This creates processing delays of several days, high operational cost, and compliance gaps when human reviewers apply policies inconsistently. Receipt fraud, duplicate submissions, and inflated claims go undetected in high-volume environments. The lack of a structured audit trail further complicates compliance reporting. There is a clear need for an intelligent, automated system capable of enforcing policy uniformly while exercising contextual judgment on ambiguous or complex claims.

- **Target Domain:** ORION targets the corporate finance and HR operations domain, specifically the employee expense management lifecycle. It is applicable to mid-to-large enterprises with high volumes of expense claims across travel, meals, accommodation, and miscellaneous business expenditure. The system is particularly relevant in regulated industries such as financial services, healthcare, and professional services, where compliance audit trails are mandatory and reimbursement fraud carries significant legal and reputational risk.

- **Proposed Solution Summary:** ORION is an AI-driven reimbursement automation platform built on a graph-based multi-agent orchestration framework using LangGraph. It combines a deterministic rule engine for non-negotiable hard policies with LLM-powered contextual intelligence for ambiguous or nuanced claims. The system ingests expense documents, validates them in parallel policy and intelligence pipelines, and routes each claim through an autonomous supervisor to an appropriate outcome: approval, clarification request, or human escalation. Every decision is logged in a persistent audit ledger with full reasoning traces, enabling complete compliance visibility.

Background & Business Objective

Background & Business Objective

- **Background of the Problem:** Traditional expense reimbursement workflows rely on manual review by finance personnel who must cross-check employee-submitted claims against physical or scanned receipts, apply policy rules from memory, and escalate edge cases to

managers. As organizations scale, this process becomes untenable. Review backlogs grow, policy application becomes inconsistent across reviewers, and the absence of machine-readable audit trails hampers internal audits and external compliance reviews. Existing ERP-integrated tools provide basic form validation but lack contextual reasoning capabilities, leaving ambiguous claims to human discretion without structured decision support.

- **Importance of solving this issue:** Unresolved, this problem compounds across three critical dimensions. Operationally, delayed reimbursements reduce employee satisfaction and add overhead cost. Financially, undetected fraud and duplicate submissions result in direct monetary loss. Compliance-wise, inconsistent enforcement exposes organizations to audit findings and regulatory penalties. Automating this process with explainable AI decisions reduces human error, ensures policy uniformity, cuts processing time by an estimated 60–70%, and provides a machine-generated audit trail that satisfies both internal governance and external regulatory requirements.

- **Strategic Fit/Impact**: ORION aligns with the enterprise-wide initiative to reduce operational overhead through intelligent automation. It supports the finance function's digital transformation roadmap by introducing a modular AI backbone that can be extended to adjacent use cases such as procurement validation and invoice processing. The system's LangSmith-integrated observability layer positions the organization to iteratively improve AI decision quality over time. Strategic impact includes reduced finance headcount for routine reviews, faster reimbursement cycles, and a defensible audit posture for regulatory engagements.

Product Purpose

Main Goal of the System

Automate the end-to-end employee expense reimbursement process with consistent policy enforcement, LLM-powered contextual reasoning, and a complete audit trail for every decision.

Intended Users (Target Audience)

- Employees: Submit expense claims with supporting documents

- Finance Reviewers: Monitor approved claims, manage policies and access the decision-making platform for review

- System Administrators: Manage configuration, health monitoring, system settings, monitor LLM performance via LangSmith and iterate on prompt & policies strategies.

System Functionalities

Description

ORION's functional architecture is organized around a graph-based workflow where a central WorkflowState object accumulates signals as the claim traverses each processing node. The system supports PDF and DOCX file ingestion, real-time parallel policy and intelligence analysis, five-way supervisor routing, and persistent decision recording.. The system is designed to operate autonomously for the majority of claims.

Key Functionalities

- Intelligent document intake: parses and extracts key information of claims from user's message and PDF and DOCX files.
- Deterministic policy engine: enforces hard rules with 100% consistency.
- LLM intelligence analysis: detects fraud signals, receipt discrepancies, and ambiguous patterns.
- Advanced deduplication: SHA256 hashing prevents double-submissions; z-score analysis flags volume anomalies.
- Parallel processing pipeline: branches into Policy Check and Intelligence nodes concurrently, reducing latency by about 40%.
- Five-way supervisor routing: Approve, Clarify, Escalate, Loop-back, or Record.
- Audit ledger (ledger.json): immutable record of all decisions with reasoning trail and LangSmith trace links.
- Health monitoring: /api/health endpoint for LLM connectivity and system status checks.

AI Model & Prompt Design

- **Model Selection:** ORION employs GPT-5.4 mini as the primary LLM for contextual intelligence tasks, including receipt cross-checking, fraud signal interpretation, and ambiguity resolution. GPT-5.4 mini is selected for its superior reasoning and instruction-following

capability on structured document inputs. Gemini is provisioned as an alternative model for specific intelligence workloads, providing redundancy and cost optimization flexibility. Deterministic policy checks are executed entirely outside the LLM to guarantee consistent, auditable outcomes.

- **Prompting Strategy:** The system uses a structured prompting approach where each LLM node receives a role-contextualized system prompt defining its scope of analysis (e.g., receipt cross-check agent, fraud detection agent). User prompts are constructed dynamically from normalized claim data and extracted receipt text. Chain-of-thought prompting is applied to encourage step-by-step reasoning, and the model is instructed to return structured JSON outputs containing findings, confidence scores, and reasoning summaries to facilitate downstream merge operations.
- **Context & Input Handling**: Claim documents are parsed by a Python-based extraction layer that converts PDF and DOCX inputs into normalized text representations. This text, combined with employee-declared metadata (amount, category, date, vendor), is injected into the LLM context window. The WorkflowState object tracks accumulated signals across nodes, ensuring that context from the Intelligence node is available to the Supervisor when making routing decisions. Token limits are managed by truncating raw receipt text while preserving all structured metadata fields.
- **Fallback & Failure Behavior:** If an LLM call fails due to rate limits, timeout, or API unavailability, the system logs the failure in the WorkflowState. Confidence scores below a defined threshold trigger a loop-back to the Intelligence node for re-analysis before escalation. If the loop-back count exceeds the maximum retry limit, the claim it is automatically escalated to a human reviewer. All failure events are captured in the audit ledger with error context for post-incident review.

User Stories & Use Cases

- **User Stories:**
    - *"*As an employee, I want to upload my expense receipts and have my claim processed automatically so that I receive reimbursement without manual delays."
    - "As a finance reviewer, I want to see only escalated claims with full AI reasoning trails so that I can make informed decisions without reviewing routine approvals by defining policies."

- “As a system administrator, I want to monitor the health of LLM connections in real time so that I can respond proactively to service degradations.”

- **Use Cases (Main Interactions):**
  - **Prompt to UI Flow:**

    UC-01: Document Upload and Submission
    - The user accesses a file upload interface supporting PDF and DOCX format.
    - Upon submission, the UI displays a processing state while the Intake Agent normalizes data.
    - The system presents a progress visualization indicating the parallel execution of Policy Check and Intelligence Analysis.
    - A final confirmation screen notifies the user of the approval status once the Recorder logs the decision.

    UC-02: Duplicate Detection Notification
    - The user attempts to submit a document previously processed.
    - The UI triggers an immediate validation overlay when the SHA256 hash match is detected.
    - The system displays a "Request Clarification" prompt, explaining the idempotency violation.
    - The flow terminates with a rejection notice, preventing further financial processing.

    UC-03: Ambiguous Claim Review
    - The Finance Reviewer logs into a specialized dashboard containing the Human Escalation queue.
    - The interface displays a detailed "AI Reasoning Summary," highlighting fraud signals and low confidence scores.
    - The UI provides manual "Approve" and "Reject" actions.
    - Once selected, the system updates the ledger with both the AI analysis and the human decision.

  - **Flow Generation:**

    Ingestion and Normalization (UC-01)
    - Input: Receipt File >>> Intake Agent.
    - Action: Transform unstructured data into a standardized schema.

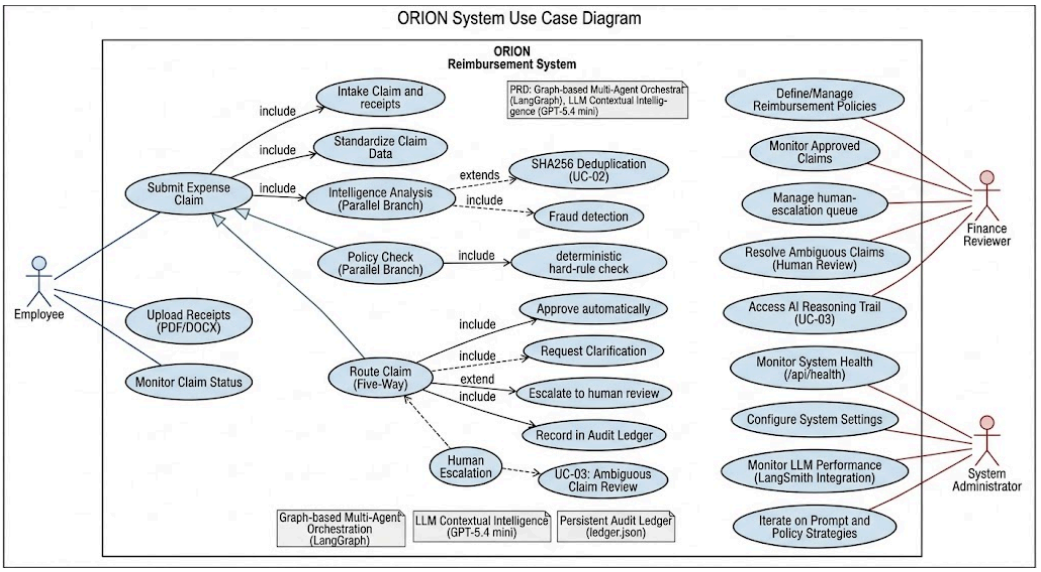
- Transition: Trigger Parallel Pipeline.

Analysis and Reconciliation (UC-01/UC-02)

- Node A (Policy Check): Validates against company spending rules.
- Node B (Intelligence Analysis): Performs SHA256 hashing and fraud detection.
- Logic Gate: If SHA256 exists in ledger.json, route to Idempotency Violation (UC-02).
- Merge Node: Reconciles outputs from Node A and Node B into a unified claim object.
- Recorder Node: Executes an append-only operation to ledger.json containing the reasoning trail.

Routing and Human-in-the-Loop (UC-03)

- Supervisor Node: Analyzes the confidence score from the Merge Node.
- Decision Branch:
  - If Confidence > Threshold: Route to Critic >> Recorder.
  - If Confidence < Threshold: Route to Human Escalation Queue.
- Recursive Loop: Allows for a finite number of automated retries before final escalation.



Features Included (Scope Definition)

- Direct document ingestion (PDF/DOCX) via specialized parsing agents.
- Deterministic hard-rule policy engine with configurable policy definitions.
- LLM-powered contextual intelligence using GPT-5.4 mini
- SHA256-based deduplication and z-score volume anomaly detection.
- Parallel workflow execution for Policy Check and Intelligence nodes.
- Five-way supervisor routing: Approve, Clarify, Escalate, Loop-back, Record.
- Critic node for final sanity check before approval recording.
- Persistent JSON audit ledger with confidence snapshots and LangSmith trace links.
- LangSmith integration for full LLM trace observability.

Features Not Included (Scope Control)

- Direct ERP or payroll system integration (e.g., SAP, Workday)
- Mobile application interface for employees.
- Real-time push notifications (email or SMS) to employees on claim status changes.
- Multi-currency conversion and foreign exchange rate handling.
- Role-based access control (RBAC) UI for managing reviewer permissions.
- OCR for handwritten receipts or non-digital image formats.
- Multi-tenant / multi-organization support in v1 architecture.
- Machine learning model fine-tuning pipeline based on historical decision data.

Assumptions & Constraints

- **LLM Cost Constraint:** ORION's LLM calls are scoped exclusively to contextual intelligence tasks; deterministic policy checks are performed by the Python engine to minimize token consumption. GPT-5.4-mini is used only where contextual reasoning is required. Each claim is expected to consume a bounded number of LLM calls (target: ≤4 per claim in standard flow). High-volume periods may require token batching strategies or model tier downgrade for non-fraud-sensitive claims to stay within monthly API budget allocations

- **Technical Constraints:**

- The system is built on LangGraph and LangChain; updates to these frameworks may require workflow refactoring.
- The JSON-based ledger is a local persistence mechanism; it is not designed for concurrent multi-instance writes without a database migration.
- Document parsing is limited to text-extractable files; scanned image-only PDFs require OCR pre-processing not included in v1.
- LangSmith integration depends on external service availability; tracing failures do not block claim processing.
- FastAPI deployment assumes a single-server setup; horizontal scaling requires session state externalization.

- **Performance Constraints:**
  - Target end-to-end processing time per standard claim: under 30 seconds.
  - Parallel branching is expected to reduce total latency by approximately 40% versus sequential processing.
  - LLM API latency is the primary bottleneck; claims with multiple loop-back retries may exceed the 30-second target.
  - The system assumes a maximum concurrent claim submission volume of 100 requests per minute in v1.

- **User Input:**
  - Employees are assumed to submit claims with at least one supporting document in a supported format.
  - Employee-declared amounts are provided in the system's base currency (multi-currency is not supported in v1).
  - Receipt text is assumed to be machine-readable; handwritten or purely image-based receipts are not supported

Risks & Questions Throughout Development

- **Risks**
  - LLM hallucination in receipt cross-check may produce false fraud flags.
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    - GPT API pricing changes could affect the operational cost model.

- **Questions during Development**
    - What is the acceptable false-positive rate for fraud escalation before employee experience degrades?
    - Should the Critic node be LLM-powered or deterministic in the production version?
    - What is the data retention policy for ledger entries and LangSmith traces?
    - How will policy rule updates be managed and versioned without redeployment?